

KHORGOTARIAT, Mongolia

WMO: 442290

Lat: 48.1567N

Lon: 99.8767E

Elev: 2181

StdP: 77.72

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.9	-31.8	-36.2	0.2	-32.9	-34.3	0.2	-30.7	9.2	-20.7	8.5	-21.5	1.6	270	0.750

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	24.0	12.0	22.2	11.4	20.6	11.0	14.4	19.7	13.3	18.7	12.5	17.7	3.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.8	12.1	15.5	11.6	11.1	14.6	10.5	10.4	13.8	48.2	19.7	44.8	18.6	42.0	17.7	18.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.3	8.3	7.2	DB	-35.8	27.3	2.7	1.7	-37.7	28.5	-39.3	29.5	-40.8	30.4	-42.8	31.6
			WB	-35.8	15.5	2.7	1.4	-37.8	16.5	-39.4	17.3	-40.9	18.1	-42.9	19.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.6	-22.6	-19.3	-9.6	0.1	5.7	11.9	14.2	12.2	6.7	-1.1	-11.3	-19.4	(6)
(7)		DBStd	13.64	5.32	5.78	6.22	5.19	4.41	3.49	2.53	3.04	3.99	4.73	5.67	5.42	(7)
(8)		HDD10.0	4920	1010	819	608	298	142	20	2	14	111	346	639	912	(8)
(9)		HDD18.3	7652	1268	1053	866	548	391	193	129	191	349	604	889	1170	(9)
(10)		CDD10.0	313	0	0	0	1	9	78	132	82	11	0	0	0	(10)
(11)		CDD18.3	3	0	0	0	0	0	1	2	1	0	0	0	0	(11)
(12)		CDH23.3	79	0	0	0	0	1	24	34	18	1	0	0	0	(12)
(13)		CDH26.7	6	0	0	0	0	0	2	2	2	0	0	0	0	(13)
(14)	Wind	WSAvg	2.9	2.6	2.8	3.3	3.8	3.6	2.7	2.3	2.2	2.6	2.9	3.0	2.8	(14)
(15)	Precipitation	PrecAvg	246	1	2	4	11	18	47	73	56	15	8	4	2	(15)
(16)		PrecMax	359	7	7	16	42	46	88	138	124	50	27	13	6	(16)
(17)		PrecMin	174	0	0	1	0	2	16	36	10	1	1	0	0	(17)
(18)		PrecStd	46	2	2	3	8	12	18	20	25	11	7	3	1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.6	-2.2	7.9	16.2	22.4	26.0	26.6	25.6	21.9	13.2	5.2	-2.9	(19)
(20)			MCWB	-7.3	-5.0	0.8	5.6	9.3	12.3	13.7	11.9	9.1	4.1	0.4	-5.5	(20)
(21)		2%	DB	-9.2	-5.4	4.9	13.3	19.5	23.3	24.1	22.6	18.5	10.5	1.4	-7.0	(21)
(22)			MCWB	-10.7	-7.5	-0.9	4.3	8.3	11.2	12.8	11.6	8.1	3.0	-2.2	-8.5	(22)
(23)		5%	DB	-12.3	-7.9	2.5	10.9	17.0	21.4	22.2	20.5	16.3	8.2	-1.2	-9.7	(23)
(24)			MCWB	-13.5	-9.7	-2.2	3.2	7.1	10.8	12.3	10.9	7.3	1.9	-3.9	-10.9	(24)
(25)		10%	DB	-15.0	-10.5	0.2	8.8	14.7	19.4	20.6	18.7	14.3	6.0	-3.5	-11.8	(25)
(26)			MCWB	-15.9	-11.9	-3.5	2.0	5.8	10.1	12.1	10.6	6.4	0.8	-5.6	-12.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.6	-4.7	1.8	6.5	11.0	14.0	16.2	15.2	10.9	4.9	0.8	-5.6	(27)
(28)			MCDWB	-6.0	-2.5	7.8	13.9	20.1	22.4	20.2	19.0	16.1	11.4	5.0	-3.1	(28)
(29)		2%	WB	-10.8	-7.4	-0.5	5.0	9.1	12.9	15.0	13.6	9.5	3.6	-2.0	-8.4	(29)
(30)			MCDWB	-9.3	-5.5	4.1	11.5	17.1	19.6	20.4	18.1	15.7	9.4	1.1	-7.0	(30)
(31)		5%	WB	-13.5	-9.6	-2.0	3.7	7.6	12.0	14.1	12.6	8.5	2.3	-3.8	-10.9	(31)
(32)			MCDWB	-12.4	-7.8	2.2	10.1	15.9	18.1	19.2	17.4	14.4	7.5	-1.4	-9.7	(32)
(33)		10%	WB	-15.8	-11.8	-3.4	2.4	6.3	11.3	13.1	11.7	7.4	0.9	-5.5	-12.9	(33)
(34)			MCDWB	-15.1	-10.5	0.2	8.1	13.6	17.3	18.0	16.4	12.8	5.3	-3.7	-11.9	(34)
(35)	Mean Daily Temperature Range	MDBR	8.2	10.6	12.8	11.4	12.1	11.6	10.8	11.1	12.1	10.2	8.8	7.3	(35)	
(36)		5% DB	MCDBR	11.3	12.8	14.7	14.7	16.9	15.4	14.7	14.9	15.5	13.4	10.8	9.4	(36)
(37)			MCWBR	10.1	10.9	9.9	7.6	7.8	5.7	5.0	5.0	6.2	6.9	7.7	8.3	(37)
(38)		5% WB	MCDWB	11.2	12.7	14.0	13.5	15.7	12.0	11.6	11.3	13.6	12.7	10.4	9.5	(38)
(39)			MCWBR	10.1	10.9	9.6	7.7	7.8	4.9	4.8	4.6	5.9	6.8	7.5	8.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.214	0.238	0.279	0.316	0.324	0.331	0.328	0.310	0.279	0.258	0.234	0.217	(40)	
(41)		taud	2.398	2.356	2.306	2.316	2.334	2.419	2.466	2.490	2.524	2.501	2.434	2.381	(41)	
(42)		Ebn at Noon	906	946	945	936	936	929	927	934	939	912	866	851	(42)	
(43)		Edh at Noon	63	84	106	117	120	112	105	98	85	73	61	57	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.84	2.92	4.42	5.68	6.42	6.34	6.01	5.35	4.56	3.06	1.90	1.46	(44)	
(45)		RadStd	0.09	0.12	0.18	0.24	0.41	0.33	0.37	0.23	0.22	0.17	0.09	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	-2.16	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page